

Project: DiscountMate

Workstream: DA-02 Insight Surfacing

Task: DA-02-T3 Document Dashboard Requirements

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Reporting period: Week 5, T2 2026

1. Purpose

-To define the dashboard requirements, user flows, navigation and expected functionality for the DiscountMate Power BI dashboard.

-The requirements are based on the existing Power BI Test Dashboard reviewed in DA-02-T1, the confirmed customer and admin/stakeholder needs, the DiscountMate Product Comparison functionality, and behavioural analytics requirements being developed by Feature 1.

-The existing dashboard will be used as the starting point rather than rebuilding the reporting solution from the beginning.

2. Current Dashboard Scope

The current Test Dashboard contains two active pages:

1. **Product Price Analysis**
2. **Category Specials**

The source PBIX metadata also contains references to **Product Comparison** and **Product Search**. However, these pages are not visible in the current published dashboard and have not been confirmed as active dashboard pages. Therefore, they are not treated as existing Power BI pages in this document.

The App Development **Product Comparison** page is still considered when defining the customer experience because the Power BI reporting should support and align with the wider DiscountMate application.

3. Stakeholder Definition

The stakeholder requirements in this document are aligned with the DA-02-T2 stakeholder dashboard views developed and reviewed by the DA-02 team. Four stakeholder groups are considered: **Customer**, **Admin**, **Mentor** and **Developer**.

The following stakeholder requirements are based on:

- DA-02-T1 existing dashboard review;

- two confirmed Power BI pages;
- DiscountMate Product Comparison functionality;
- agreed customer and admin/stakeholder requirements;
- Feature 1 behavioural analytics requirements.

Four stakeholder groups are considered.

3.1 Customer / Shopper

Purpose: Help customers discover products and make better purchasing decisions.

Key information:

- product search,
- categories,
- product details,
- prices across stores,
- savings/value indicators,
- filters,
- favourites and product actions.

3.2 Admin

Purpose: high-level view of platform activity and product/store data.

Key information:

- user activity
- product catalogue
- product categories
- store/product availability
- popular products
- analytics summaries
- data quality/status
- important alerts

3.3 Mentor

Purpose: high-level overview of project progress and analytics/ML outcomes.

Key information:

- overall project progress
- analytics/ML results
- data quality
- refresh status
- price/discount trends
- customer insights/personas
- current issues/blockers

3.4 Developer

Purpose: technical overview of the DiscountMate system.

Key information:

- API/service health
- API errors/failures
- system performance
- data-processing status
- backend–analytics integration
- refresh status
- technical issues

4. Dashboard Requirements

4.1 Customer / Shopper Requirements

The customer-facing dashboard should focus on simple and useful information that helps users make price-comparison decisions.

ID	Requirement	Priority
C1	Allow customers to select a product or category.	High
C2	Show current product prices by retailer.	High
C3	Clearly identify the lowest or best available price.	High
C4	Show product price history over time.	High
C5	Show current specials and discount percentages.	High
C6	Allow users to filter specials by category and discount range.	High

C7	Display useful product information such as brand, product name, pack/unit and retailer.	High
C8	Show potential savings and relevant price metrics where available.	Medium
C9	Explain discount and savings measures in simple language.	Medium
C10	Keep dashboard design consistent with the DiscountMate Product Comparison experience.	Medium

The customer view should avoid requiring users to interpret several complex tables before identifying the best option. Important information such as the **lowest price, strongest special and retailer comparison** should be visually clear.

4.2 Admin Dashboard Requirements

ID	Requirement	Priority
A1	Show product price trends over time.	High
A2	Allow comparison of pricing across retailers.	High
A3	Show promotional and discount activity.	High
A4	Allow analysis by product, category, retailer and reporting period.	High
A5	Show important price and promotion measures already available in the model.	High
A6	Show data coverage and latest refresh information where available.	Medium
A7	Support identification of popular or frequently compared products when behavioural data becomes available.	Medium
A8	Support behavioural/persona insights when Feature 1 outputs become available.	Medium
A9	Make data-quality or missing-data issues visible where appropriate.	Medium
A10	Reuse existing Power BI calculations and model fields where suitable.	High

4.3 Mentor Dashboard Requirements

ID	Requirement	Priority
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M1	Show overall project progress.	High
M2	Show key analytics and ML outcomes.	High
M3	Show data quality and refresh status.	High
M4	Show key price and discount trends.	Medium
M5	Show customer insights/personas when available.	Medium
M6	Show current project issues or blockers.	High

4.4 Developer Dashboard Requirements

ID	Requirement	Priority
D1	Show API/service health.	High
D2	Show API errors and failures.	High
D3	Show system/service performance.	High
D4	Show data-processing status.	High
D5	Show backend–analytics integration status.	High
D6	Show data/service refresh status.	Medium
D7	Show current technical issues.	High

The current Product Price Analysis and Category Specials pages mainly support the Customer view. Future stakeholder-specific views can extend this reporting structure for Admin, Mentor and Developer needs.

5. Requirements for Existing Dashboard Pages

5.1 Product Price Analysis

The existing **Product Price Analysis** page should be retained and improved.

Required functionality

The page should:

- allow category selection;

- allow product selection;
- compare prices between available retailers;
- display product price history by date and retailer;
- display the product image where available;
- show lowest price;
- show highest price;
- show average price;
- show maximum discount;
- show average, maximum and total discounted days;
- show the number of discount events; and
- show percentage of time on special.

Recommended improvements

The page should make the current best price or cheapest retailer more prominent, so users do not need to interpret multiple metrics to find the best option.

The existing discount calculations should also be explained in simple language so customers can understand what measures such as discounted days and percentage on special represent.

6. Category Specials Requirements

The existing **Category Specials** page should also be retained and improved.

Required functionality

The page should:

- allow users to select a category;
- allow users to select a discount range;
- allow users to display lowest prices only;
- show product brand;
- show product name;
- show pack/unit information;
- show current price;
- show discount percentage;
- show retailer;
- show date on special;
- indicate whether the current price is the lowest price; and
- show the best recorded price where available.

Recommended improvements

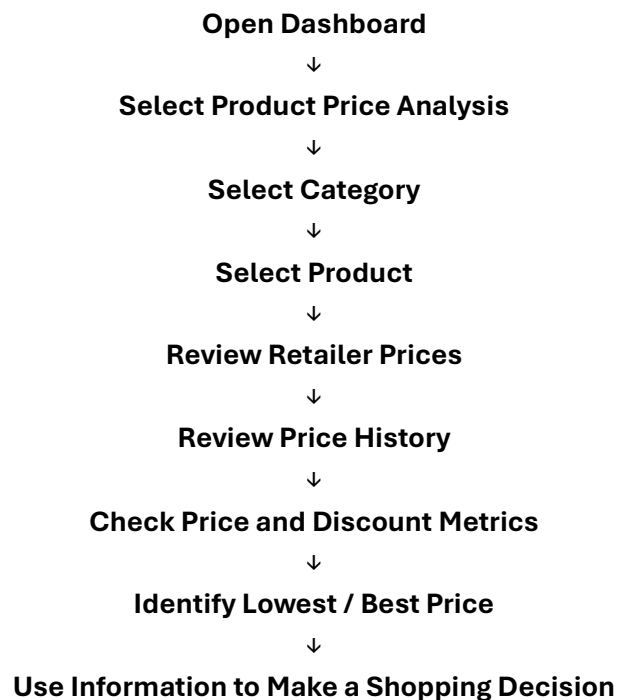
The strongest specials should be easier to identify visually.

The existing detailed table is useful, but long product names can make the page difficult to scan. Important fields such as current price, discount percentage, retailer and best price should therefore remain visually prominent.

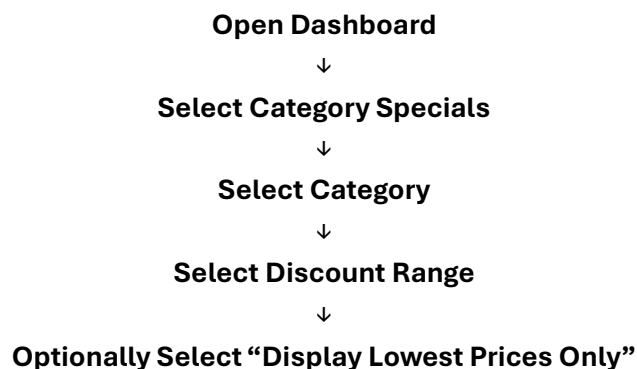
7. User Flow

7.1 Customer User Flow

Flow 1: Product Price Comparison



Flow 2: Finding Specials





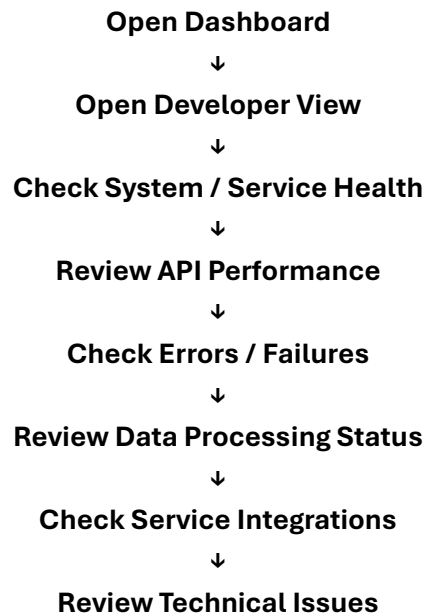
7.2 Admin User Flow



7.3 Mentor User Flow



7.4 Developer User Flow



8. Dashboard Navigation

Future Stakeholder Navigation

Based on the stakeholder requirements defined in DA-02-T2, the dashboard can be extended to provide stakeholder-specific views. The proposed navigation structure is:

DiscountMate Dashboard

→ Customer Dashboard

- Product Price Analysis
- Category Specials

→ Admin Dashboard

- Platform & Management Overview

→ Mentor Dashboard

- Project & Analytics Overview

→ Developer Dashboard

- Technical & System Overview

The Customer Dashboard can continue to use the existing Product Price Analysis and Category Specials pages. The Admin, Mentor and Developer dashboards are proposed future stakeholder-specific views based on the DA-02-T2 requirements. These views should be implemented when the required data and technical support become available.

9. Data and Functional Requirements

The existing dashboard already uses information relating to:

- product;
- category;
- retailer;
- product image;
- date;
- current price;
- lowest/best price;
- unit price;
- discount percentage;
- promotion status; and
- price history.

The existing model also contains calculations supporting price and promotion analysis.

Where suitable, existing model fields, transformations and DAX calculations should be reused rather than recreated.

Future behavioural dashboard elements will depend on the Feature 1 output. These may include behavioural segments/personas and popular or frequently compared products.

The dashboard should also display the **latest data refresh information** where technically available so internal users understand how current the displayed information is.

9.1 Existing PBIX and DAX Review

The existing Power BI files were considered when defining these requirements to avoid duplicating fields and calculations already available in the current model.

Files reviewed:

- Test_Dashboard.pbix
- DAX_Columns.xlsx
- DAX_Queries.xlsx
- Power BI MongoDB Connection Guide

The existing model already supports several requirements identified in this document, including:

- minimum, maximum and average price;
- discount amount and percentage;
- price change calculations;

- average, maximum and total days on special;
- discount event counts;
- percentage of time on special;
- lowest/best price indicators;
- retailer name standardisation;
- product ID cleaning;
- numeric unit-price conversion; and
- previous price/date calculations.

These existing fields, measures and transformations should be reused where appropriate rather than recreated. New calculations should only be proposed where an identified stakeholder requirement cannot be met using the existing model.

Future Gold layer requirements

For future dashboard development, the Data Engineering team has also been asked to support:

- a clean product pricing fact table;
- stable product, retailer and category keys;
- numeric price and unit-price fields;
- previous price and price-change fields;
- discount start and end dates;
- days-on-special measures;
- discount event identifiers;
- savings measures;
- best-retailer indicators;
- data-quality flags; and
- refresh timestamps.

These requirements support the customer and internal stakeholder views defined above and should be reused when they become available through the Gold layer.

10. Design and Usability Requirements

The dashboard should:

- use clear and consistent terminology;
- maintain consistent colours, fonts and cards across pages;
- align visually with the wider DiscountMate application where practical;
- make important information easy to identify;
- avoid unnecessary visual complexity;
- provide clear filter labels;

- clearly highlight lowest prices and strong specials;
- explain unfamiliar calculations where necessary; and
- separate customer-focused insights from detailed internal analysis where appropriate.

The dashboard should improve the existing work rather than replacing functioning components without a clear requirement.

Customer views should remain simple and decision-focused, while Admin, Mentor and Developer views may contain more detailed operational or technical information.

11. Dependencies and Limitations

The following dependencies should be considered during development: **DA-02-T1**

The existing dashboard review provides the baseline for the requirements in this document.

DA-02-T2

The stakeholder views defined in DA-02-T2 are now used as the main stakeholder reference for this document. The four agreed stakeholder groups are Customer, Admin, Mentor and Developer. The requirements and user flows in T3 have been updated to align with these stakeholder views.

Feature 1 Behavioural Analytics

Behavioural/persona dashboard elements depend on the behavioural feature and clustering outputs being produced by Feature 1.

Existing Power BI Model

The existing PBIX/DAX work should be reused where appropriate. Development should confirm that the required measures and fields remain suitable before implementation.

12. Recommended Dashboard Structure

Based on DA-02-T2, four stakeholder-specific dashboard views are recommended:

- **Customer Dashboard:** Product discovery, price comparison, specials and savings.
- **Admin Dashboard:** Platform activity, product/store data, data quality and alerts.
- **Mentor Dashboard:** Project progress, analytics/ML results, trends, insights and blockers.
- **Developer Dashboard:** System health, API performance, errors, data processing and integrations.

The existing Product Price Analysis and Category Specials pages will support the Customer Dashboard. Other stakeholder views can be developed when required data becomes available.

13. Conclusion

The existing DiscountMate Power BI dashboard provides a strong foundation for future development. Based on **DA-02-T2**, the dashboard should support **Customer, Admin, Mentor** and **Developer** views while reusing existing functionality where possible. Further improvements can be implemented as the required data becomes available.